

Thermomechanical Stress-Aware Management for 3-D IC Designs

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Abstract—Thermal characteristics have been considered as one of the most challenging problems in 3-D integrated circuits (3-D ICs). Due to the thermal expansion coefficient mismatch between through-silicon vias (TSVs) and the silicon substrate, and the presence of elevated thermal gradients, thermomechanical stress issues are exacerbated in 3-D ICs. In this brief, we propose a solution that combines design-time and run-time techniques to reduce thermomechanical stress and the associated reliability issues. A TSV stress-aware floorplan policy is proposed to minimize the possibility of wafer cracking and interfacial delamination. In addition, a run-time thermal management scheme effectively eliminates large thermal gradients between layers. Experimental results show that the reliability of 3-D design can be significantly improved due to the reduced TSV thermal load and the elimination of mechanical damaging thermal cycling pattern. Moreover, impacts of thermal characteristics in TSVs and thermal vias insertion are explored.

Index Terms—3-D integrated circuits (3-D ICs), mechanical reliability, thermal cycling, thermal management, thermomechanical stress.

I. INTRODUCTION

The 3-D integrated circuits (3-D ICs) have been shown to provide numerous advantages in achieving better performance for future computing systems [1], [2]. However, the increased power density and decreased chip footprint induce thermal issue, which is widely considered as one of the biggest challenges in 3-D designs [3], [4].

In addition to temperature increases, the coefficient of thermal expansion (CTE) mismatch exacerbates thermal stresses in 3-D circuits. Merely using design-time solution is insufficient toward thermal stress problem, because the dynamic thermal gradient cannot be captured. To this end, 3-D IC designs should carefully take into account the aforementioned thermal challenges in both design-time and run-time. Unfortunately, little work has been done to alleviate the challenge through these two stages. In this brief, we propose a two-stage, design-time and run-time solution to this problem. The proposed two-stage thermal management methodology can be summarized as follows.

- 1) Design-time thermal stress-aware floorplan to reduce the through-silicon via (TSV) thermal load in addition to the whole chip temperature and minimize thermomechanical stresses.
- 2) Run-time thermal management scheme to analyze thermal cycling patterns and control thermal gradients, thus to lower mechanical stresses on chip and eventually achieve mechanical equilibrium among layers.

II. BACKGROUND AND RELATED WORK

Stacked chips on 3-D architecture increase the packaging density and thermal resistances, which results in higher on-chip

temperatures. Extensive studies have been done by focusing on the 3-D thermal modeling, analysis [5], [6], and thermal-aware design methodology [7], [8] to manage the on-chip thermal issues of 3-D ICs. However, the TSV lateral thermal blockage effect and the silicon fatigue due to thermomechanical stresses were not considered in previous work. On the other hand, prior work on analyzing the mechanical stresses in 3-D ICs [9], [10] only consider the static stress management by adjusting TSV keep-out zone size, TSV placement, or TSV structure. Distinguished from previous work, this brief not only accounts for the static (design-time) management of thermal stress and thermal load but also takes into account the run-time stress analysis and management.

A. Analysis of TSV Thermal Effects

To minimize the chip temperature and thermomechanical stresses-induced fatigue, the placement of TSV farms can be optimized during design time. Two of the TSV thermal effects should be considered: TSV thermomechanical stresses and lateral thermal blockage effect.

1) *TSV Thermomechanical Stresses*: In 3-D IC fabrication, copper (Cu) is usually used as TSV filling material that has more than five times larger CTE than silicon. The CTE mismatch between TSV and silicon substrate in turn introduces mechanical stresses that can lead to high probability of die cracking and interfacial delamination [11], [12]. Their study suggested that the stresses in TSVs are proportional to the thermal load and CTE mismatch. And the stresses can be represented by radial σ_r , circumferential σ_θ , and axial stresses σ_z , which has the following expression:

$$\sigma_r = \sigma_\theta = \frac{-E(\alpha_{\text{tsv}} - \alpha_{\text{si}})T_{\text{tsv}}}{2 - 2\nu}, \quad \sigma_z = 2\sigma_\theta \quad (1)$$

where α_{tsv} is the CTE of TSVs and α_{si} denotes the CTE of silicon. T_{tsv} is the thermal load on TSVs, which can be calculated as the difference between TSV temperature and room temperature. E is Young's modulus and ν is Poisson's ratio.

When the material and diameter of TSVs are determined, and the other parameters are constant, and therefore, the only variable is TSVs' thermal load. Therefore, the proposed design-time thermal management scheme alleviates thermomechanical stresses by reducing the thermal load on TSVs.

2) *TSV Lateral Thermal Blockage Effect*: TSV lateral thermal blockage effect is found due to the unequal thermal conductivities in vertical and horizontal directions [6]. Specifically, the liner layer between TSV filling material and silicon substrate has higher thermal resistance. Therefore, the lateral thermal dissipation path is blocked by liner layers. The thermal dissipation capability of TSV can be modeled with three factors: thickness of the material, thermal conductivity of the corresponding material per unit volume, and the area of the corresponding material. Then, the lateral thermal resistance can be calculated from the thermal resistance of TSV filling, liner materials, and silicon substrate [13].

B. 3-D Thermal Cycling Effect

Thermal cycling effect is another factor that can cause thermomechanical issues in 3-D ICs [14]. Fig. 1 shows that the generated thermal expansion forces are highlighted by arrows, pointing from

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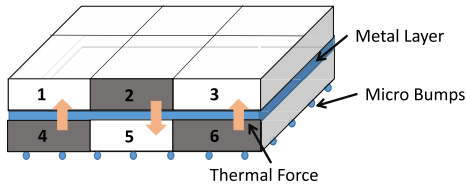


Fig. 1. Stack level thermal cycling effect in 3-D structure. Thermal stresses are pointing from hot blocks (dark color) to cool blocks (light color). Alternating direction of stresses (the arrows) easily cause fatigue or even cracking on thinned substrate.

the hotter blocks to the cooler blocks. As shown, the temperature in blocks 2, 4, and 6 is higher than that of blocks 1, 3, and 5. The corresponding forces labeled 1 and 3 are in the opposite direction with force 2. Forces in opposite directions cause fatigue in the stacked chips and make the thinned silicon substrate more vulnerable to be cracked. A run-time thermal cycling management scheme is proposed to eliminate the damaging thermal cycling pattern by using dynamic power scaling.

III. THERMOMECHANICAL STRESS-AWARE 3-D DESIGN METHODOLOGY

A. Thermomechanical Stress Aware Design Time Algorithm

The purpose of the design-time thermal stress-aware TSV management aims at reducing the CTE-induced thermomechanical stress, which can be translated into the minimization of the TSVs thermal load and chip temperature.

Intuitively, placing TSVs away from hot regions can reduce their thermal load. However, in most scenarios, hotspots are functional units that are frequently activated and highly utilized, indicating high requirements for connectivity. Timing/performance and other design goals prevent moving TSV farms away from those hotspots. As a result, the thermal stress-aware floorplan is obligatory to sustain the high circuit performance without causing severe thermal reliability problems.

In addition to the traditional floorplan solutions that make great efforts on balancing the area and performance tradeoff, the proposed floorplan flow also strives to minimize the thermal-induced stresses. Circuit description and average power consumption of each block are given as inputs. The circuit description consists of block descriptions and the connectivity information. The average power consumption of each block is then estimated from the power density on the chip. TSV farms are treated as soft blocks in the floorplan with given thermal characteristics described in Section II.

A simulated annealing-based floorplanner is employed in the flow along with an analytical initial floorplan to speed up the convergence. The circuit is partitioned into required tiers by balancing the TSV number and chip area. Then, the initial floorplan is performed analytically by placing the modules that have low power density around TSV farm. The TSV and tier temperature are generated correspondingly. After the initial floorplan, modules are randomly selected, permuted, and adjusted aspect ratio to obtain a better thermal distribution across the whole chip. Changing the aspect ratio of TSV farm means adjusting the arrangement of TSVs, resulting in better lateral thermal path for better heat dissipation.

A cost function is associated considering the chip area, average temperature, TSV farms' temperature, and wire length. After each iteration, the cost is calculated based on the floorplan thermal profile to guide the floorplan. The iteration process terminates once the SA convergence condition is satisfied or maximum iteration step is reached.

B. Thermal Cycling-Aware Run-Time Management

Run-time thermal cycling management scheme is devised as the second stage to achieve 3-D architecture mechanical equilibrium by eliminating mechanical damaging cycling patterns as illustrated in Section II.

The run-time management is performed in a bottom-up, layer-by-layer fashion. Each tier is partitioned in fine-granularity, and the temperature of each grid is monitored by thermal sensors in runtime. If sensors are not applicable, the thermal profile can be obtained with the power trace derived from the supply voltage and activity factor of each block.

The first step controls the temperature gradients of each grid to eliminate large temperature gradients between adjacent grids.¹ Correspondingly, mechanical force vectors are generated based on these adjacent temperature gradients. On the other hand, predefined thresholds are determined based on the TSV size, material, and substrate thickness. The force vectors is then compared with the predefined thresholds; if force vectors are larger than thresholds, dynamic power scaling techniques, such as DVFS, will be deployed to control the thermal dissipation of hot grids.

In addition to temperature gradients, the thermal cycling pattern is another dynamic parameter that should be handled carefully. The cycling pattern is taken into account by comparing force vectors of neighboring grids in two adjacent layers. If the thermal cycling pattern is in an alternating way as described in Fig. 1, power management is applied to the higher temperature regions to lower the resulting thermal mechanical stresses and achieve mechanical equilibrium. The power scaling results in a new thermal cycling pattern in the stack, which may require further adjustment in the next sampling interval until the whole stack reaches the mechanical equilibrium state.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

To evaluate the proposed thermal management mechanism, a typical 3-D floorplanner (3DFP) [15] that aims at reducing the average on-chip temperature is employed as our baseline. We successfully implemented the thermal stress-aware floorplan mechanism in 3DFP. We also extended Hotspot [16] to estimate 3-D on-chip temperature with detailed modeling on TSV lateral and vertical thermal conductivities. MCNC benchmarks and one 3-D stacking system with four cores and shared L2 cache are leveraged in the block-level and system-level evaluations. We differentiated the block-level simulation and the system-level simulation, since they have distinct thermal characteristics. The temperature gradients at block level are larger and more unpredictable due to various features of the underlying circuits, while the temperature distribution at system level is usually uniform and predictable because of the relatively regular placement of functional modules.

The thermal parameters used in the experiment are listed in Table I. The default TSV diameter is 10 μm . Since the TSV farm size and arrangement are different in each circuit for block-level evaluation, we calculated the corresponding vertical and lateral thermal resistivities with the model proposed in [13] and used them as input parameters for the floorplan. In addition to the mechanism evaluation, we varied the TSV lateral thermal conductivity and TSV diameter for the sensitivity study. For simplicity, the elastic mismatch between silicon and copper is neglected.

A. Block-Level Thermomechanical Stress-Aware Floorplan

To quantify the TSV thermal load reduction from the stress-aware floorplan, we conducted experiments on five MCNC floorplan

¹In this brief, the temperature differences between grids are used to represent temperature gradients for simplicity.

TABLE I
THERMAL PARAMETERS THAT ARE USED IN THE EXPERIMENTS;
SENSITIVITY STUDY IS PERFORMED WITH VARIOUS TSV
LATERAL THERMAL CONDUCTIVITY AND TSV DIAMETER

silicon thickness	100 μm
silicon thermal conductivity	149 W/(mK)
silicon heat capacity	1.75×10^6 J/(m ³ K)
thermal interface material (TIM) thickness	20 μm
TIM thermal conductivity	4 W/(mK)
TIM heat capacity	4×10^6 J/(m ³ K)
TSV lateral thermal conductivity	100 - 200 W/(mK)
Thermal Vias vertical thermal conductivity	350 W/(mK)
heat sink thermal conductivity	200 W/(mK)
TSV diameter	5 - 10 μm

benchmarks: ami33, ami49, hp, xerox, and apte. These benchmarks are duplicated into two tiers for simplicity. Note that higher level of stacking can be supported with our framework. The average power density for each circuit is within the range of 0.5–2.4 W/mm². TSV farms are created based on the connectivity information between blocks.

The experimental results are shown in Table II. As shown in Table II, the maximum and average ratio of TSV temperature reduction are 19.24% and 7.56%, respectively. The peak on-chip temperatures have been reduced for most benchmarks after considering the TSV lateral thermal conduction. The peak temperatures increased in hp and xerox, because two local hotspots are directly contacted after moving the TSV farm away. TSVs with high lateral thermal resistance can also be used as a shield to separate two hotspots. The ratios of TSV axial thermal stress reduction are represented as the percentage of temperature reduction, and they are listed in the last column. By moving the TSV farm away from hotspots, the TSV temperature can be reduced with slight area overhead.

B. System-Level Design-Time and Run-Time Thermal Management Scheme

In addition to the block-level simulation, we also applied the thermal management on a system-level 3-D design. The 3-D system contains three tiers in a fashion of face-to-back bonding. The top layer mimics a multicore processor, where four SPARC-like cores with private L1 cache are deployed. And the heat sink is directly attached to the top layer. The two bottom layers are shared L2 cache. TSV buses are integrated as connection between core layer and the two cache layers. To guarantee the bandwidth, each core owns one TSV bus. The design-time mechanism generated the floorplan of core as shown in Fig. 2. The overall temperature on TSV bus is decreased compared with the floorplan with TSV buses placed in the vertical direction. In general, the stress-aware floorplan can decrease the average temperature on TSV bus by 2.15%.

In the run-time thermal management scheme, we leverage accurate thermal information provided by HotSpot in grid granularity for temperature evaluation. As shown in Fig. 3, at time 0, the top core layer has larger on-chip temperature with four hotspots in functional units *IntReg* while half part of L2 cache is in low temperature status, introducing thermomechanical stresses pointing from hotspots in core layer to the cooler half of cache. Then, at time 1, two functional units in the two bottom cores (core 0 and core 1) have lower power consumption and became cooler. However, the temperature of L2 cache increased due to the higher utilization. Then, the thermomechanical stresses are from cache layer to the core layer. This kind of alternating thermal stresses would easily cause silicon fatigue, or even result in cracking in the thinned substrate. After the power throttling on the

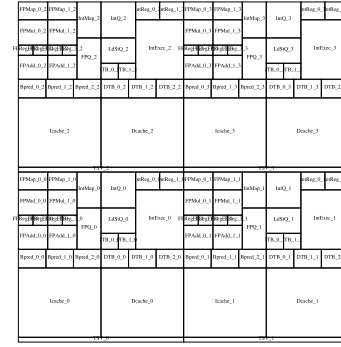


Fig. 2. Floorplan of four cores with private L1 cache.

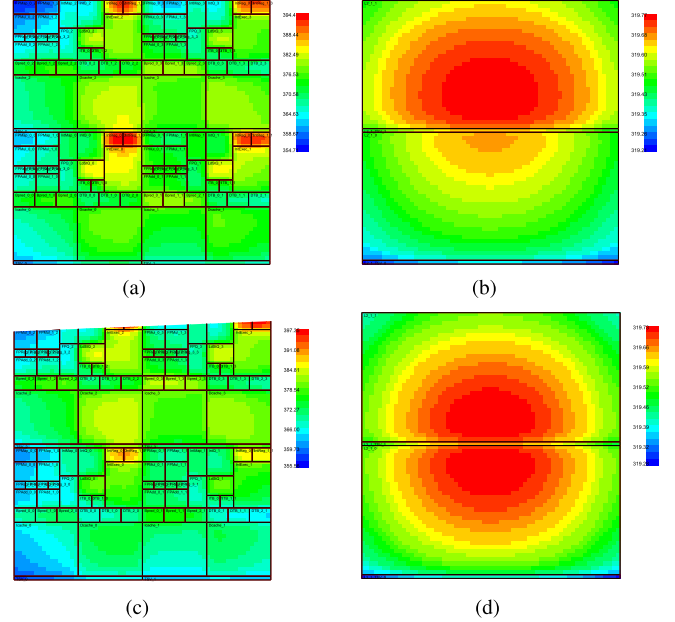


Fig. 3. Thermal cycling effects on stacked core and cache. (a) Core temperature distribution at time 0. (b) Stacked L2 cache temperature distribution at time 0. (c) Core temperature distribution at time 1. (d) Stacked L2 cache temperature distribution at time 1.

corresponding functional units in cores to reduce the thermal gradient between cores and stacked cache at time 0, the temperature of this region is decreased and the mechanical damaging thermal cycling pattern is eliminated.

C. Sensitivity Study on TSV Thermal Conductivity and Diameter

With technology scaling, smaller TSVs and advanced materials will be applied. Therefore, the influence of different thermal conductivities should be studied. In this section, we took benchmark hp as an example and showed the sensitivity of average temperature in different configurations. According to the compact thermal model [13], the lateral and vertical thermal resistivities of TSV are related to the arrangement of TSV array. Therefore, we assumed that the TSV array is arranged in square for simplicity. In order to perform a fair comparison, we used the same floorplan when the TSV size is changed. We further assumed that the chip area occupied by TSV farm keeps the same. Therefore, the decreased TSV size means larger area of silicon space between two TSVs. From the calculation, with smaller TSV size, the lateral thermal resistivity is decreased, while the vertical thermal resistivity is slightly increased. The TSV diameter is first fixed as 10 μm and the thermal conductivity increases at the step of 20 W/(mK). Then, we fixed the TSV lateral thermal conductivity as 159 W/(mK) and increase the TSV diameter

TABLE II
DESIGN TIME THERMOMECHANICAL STRESS-AWARE FLOORPLAN EXPERIMENTAL RESULTS

Circuit	Baseline			Proposed			Stress Reduction (MPa)
	avg T (K)	peak T (K)	TSV T (K)	avg T (K)	peak T (K)	TSV T (K)	
ami33	329.224	338.61	322.9	321.75	333.87	313.33	19.24 %
ami49	379.90	433.61	349.42	371.41	400.27	348.85	0.7 %
apte	370.338	418.20	352.11	369.30	415.69	349.33	3.52 %
hp	338.69	351.47	327.01	328.75	352.67	324.88	3.95 %
xerox	355.24	372.88	339.43	355.63	376.31	332.52	10.41 %

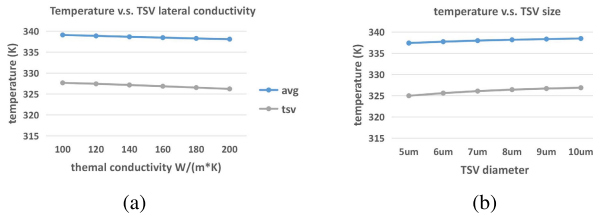


Fig. 4. TSV thermal conductivity and diameter sensitivity study results. (a) Temperatures with different TSV thermal conductivity settings. (b) Temperature results with different TSV diameters.

from 5 to 10 μm with incremental step as 1 μm . Fig. 4 shows the temperature results of different TSV thermal conductivities and diameters.

With the increased TSV thermal conductivity, the TSV temperature is slightly reduced [Fig. 4(a)]. The TSV temperature is influenced by both its own thermal features and the whole chip temperature. When the TSV power consumption, silicon to TSV ratio, and chip area occupied by TSV keep constant, only changing the TSV thermal conductivity will not have much impact on its temperature.

When increasing the TSV size, the lateral TSV conductivity decreases from 302.57 to 159.12 W/(mK). This is because with larger TSV size, the area occupation of silicon becomes smaller, and the resistance of the silicon part increases, while TSV's resistance keeps the same. Larger thermal resistance represents smaller thermal conductivity. The vertical TSV conductivity has slightly increased from 148.81 to 170.09 W/(mK) due to larger cross-sectional area of TSV filling. Therefore, in Fig. 4(b), results show that when we increase TSV diameter from 5 to 10 μm , the temperature increased from 324.97 to 326.85 K. The average on-chip temperature is only slightly increased.

D. Thermal Through-Silicon Vias Effects

In order to handle the thermal issues, thermal vias are usually inserted to help building efficient vertical thermal dissipation path. Thermal vias are the vertical metal connections that only serve as heat removal without transmitting any signals [17]. To illustrate the effect of thermal via and the impact of thermal via insertion on the floorplan, we take benchmark *hp* as an example. All configurations are consistent with values in Table I. Since there is no signal transmission in thermal vias, we assumed that thermal via has no power consumption. The generated thermal maps of floorplans with and without thermal vias are shown in Fig. 5.

When thermal vias are inserted, the floorplan has higher aspect ratio to gain the optimal on-chip and TSV temperature. We can see from Fig. 5(a), component *ppd* is the local hotspot due to the relatively larger power density. After inserting thermal vias, the peak temperature on chip drops from 354.03 to 352.48 K as shown in Fig. 5. By trying to reduce the average on-chip temperature, the thermal vias are inserted near the hotspot *ppd*.

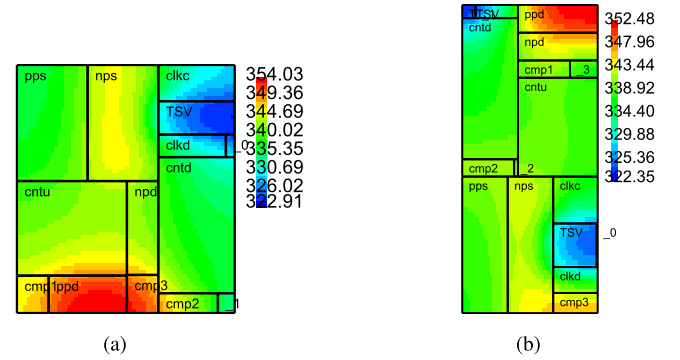


Fig. 5. Thermal maps of benchmark *hp* with and without thermal vias. (a) Without thermal vias. (b) With thermal vias.

V. CONCLUSION

This brief presents a two-stage thermal management technique on design time and run time to alleviate the thermal challenges on 3-D architectures. The results show that design-time floorplan can effectively reduce TSV thermal load. The maximum and average thermal stress reduction percentages are 19.24% and 7.56% in block level, respectively. Even in the system-level with relatively regular placement, the reduction percentage can be 2.15%. The core to memory stacking can achieve mechanical equilibrium through dynamic power scaling with slightly performance overhead. The temperature variation with different thermal conductivities and TSV diameters, and thermal via insertion are also studied.

REFERENCES

- [1] G. H. Loh, Y. Xie, and B. Black, "Processor design in 3D die-stacking technologies," *IEEE Micro*, vol. 27, no. 3, pp. 31–48, Jun. 2007.
- [2] Y. Xie, G. H. Loh, B. Black, and K. Bernstein, "Design space exploration for 3D architectures," *J. Emerg. Technol. Comput. Syst.*, vol. 2, no. 2, pp. 65–103, Apr. 2006.
- [3] P. Leduca *et al.*, "Challenges for 3D IC integration: Bonding quality and thermal management," in *Proc. IEEE Int. Interconnect Technol. Conf.*, Jun. 2007, pp. 210–212.
- [4] K. Puttaswamy and G. H. Loh, "Thermal analysis of a 3D die-stacked high-performance microprocessor," in *Proc. ACM Great Lakes Symp. VLSI*, 2006, pp. 19–24.
- [5] F. Wang, Z. Zhu, Y. Yang, and N. Wang, "A thermal model for the top layer of 3D integrated circuits considering through silicon vias," in *Proc. IEEE Int. Conf. ASIC*, Oct. 2011, pp. 618–620.
- [6] Y. Chen, E. Kursun, D. Motschman, C. Johnson, and Y. Xie, "Analysis and mitigation of lateral thermal blockage effect of through-silicon-via in 3D IC designs," in *Proc. IEEE/ACM Int. Symp. Low-Power Electron. Design*, Aug. 2011, pp. 397–402.
- [7] J. Cong, G. Luo, J. Wei, and Y. Zhang, "Thermal-aware 3D IC placement via transformation," in *Proc. Asia South Pacific Design Autom. Conf.*, 2007, pp. 780–785.
- [8] J. Cong, G. Luo, and Y. Shi, "Thermal-aware cell and through-silicon-via co-placement for 3D ICs," in *Proc. ACM/EDAC/IEEE Design Autom. Conf.*, Jun. 2011, pp. 670–675.
- [9] K. Athikulwongse, A. Chakraborty, J.-S. Yang, D. Pan, and S. K. Lim, "Stress-driven 3D-IC placement with TSV keep-out zone and regularity study," in *Proc. IEEE/ACM Int. Conf. Auton. Comput.*, 2010, pp. 669–674.

- [10] M. Jung, J. Mitra, D. Z. Pan, and S. K. Lim, "TSV stress-aware full-chip mechanical reliability analysis and optimization for 3D IC," in *Proc. ACM/EDAC/IEEE Design Autom. Conf.*, Jun. 2011, pp. 188–193.
- [11] K.-H. Lu, S.-K. Ryu, J. Im, R. Huang, and P. S. Ho, "Thermomechanical reliability of through-silicon vias in 3D interconnects," in *Proc. IEEE Int. Rel. Phys. Symp.*, Apr. 2011, pp. 3D.1.1–3D.1.7.
- [12] S.-K. Ryu, K.-H. Lu, X. Zhang, J.-H. Im, P. S. Ho, and R. Huang, "Impact of near-surface thermal stresses on interfacial reliability of through-silicon vias for 3-D interconnects," *IEEE Trans. Device Mater. Rel.*, vol. 11, no. 1, pp. 35–43, Mar. 2011.
- [13] Z. Liu, S. Swarup, S. X.-D. Tan, H.-B. Chen, and H. Wang, "Compact lateral thermal resistance model of TSVs for fast finite-difference based thermal analysis of 3-D stacked ICs," *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 33, no. 10, pp. 1490–1502, Oct. 2014.
- [14] C. Noritake, P. Limaye, M. Gonzalez, and B. Vandevelde, "Thermal cycle reliability of 3D chip stacked package using Pb-free solder bumps: Parameter study by FEM analysis," in *Proc. Int. Conf. Thermal, Mech. Multiphys. Simulation Experim. Micro-Electron. Micro-Syst.*, Apr. 2006, pp. 1–6.
- [15] W.-L. Hung, G. M. Link, Y. Xie, N. Vijaykrishnan, and M. J. Irwin, "Interconnect and thermal-aware floorplanning for 3D microprocessors," in *Proc. Int. Symp. Quality Electron. Design*, 2006, pp. 98–104.
- [16] W. Huang, S. Ghosh, S. Velusamy, K. Sankaranarayanan, K. Skadron, and M. R. Stan, "HotSpot: A compact thermal modeling methodology for early-stage VLSI design," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 14, no. 5, pp. 501–513, May 2006.
- [17] B. Goplen and S. Sapatnekar, "Thermal via placement in 3d ics," in *Proc. ACM Int. Symp. Phys. Design*, 2005, pp. 167–174.